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Postgraduate Course:

103: Probabilistic Analysis of Electrical Systems

Wasserstein-Based Control Architecture:
From agent dynamics to optimal
transport regulation

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Abstract

Wasserstein distance is a metric defined on a metric space that measures the distance between two probability distributions, either discrete or continuous. It is commonly characterized through the optimal transport formulation, which provides a principled way to compute the minimum cost required to transform one distribution into another.

In this work, the Wasserstein distance is employed for the optimal navigation of a swarm of mobile robots. The swarm configuration is described as a probability distribution characterized by its mean and covariance. A Wasserstein-based control strategy is developed to regulate these statistical quantities toward prescribed desired values, ensuring that the collective behavior of the agents converges to a target Gaussian distribution in the optimal transport sense. The resulting control input is then translated into linear and angular velocity commands for each individual agent, while respecting the nonholonomic dynamics of wheeled mobile robots. This approach enables cooperative navigation of the swarm toward a common objective in an optimal manner.

1 Introduction

The control of large-scale multi-agent systems has received considerable attention due to its relevance in applications such as robotic swarms, autonomous systems, and networked control. Classical approaches often rely on pointwise trajectory tracking or agent-level coordination laws, which may become inefficient or difficult to scale as the number of agents increases.

Recently, distributional and density-based control frameworks have emerged as a powerful alternative, where the collective behavior of the system is modeled through probability distributions rather than individual states [1, 2]. Within this paradigm, optimal transport theory provides a principled mathematical foundation for comparing and steering probability measures, with the Wasserstein distance playing a central role [3].

The Wasserstein distance has been successfully employed in control and optimization problems, including minimum-energy density steering of linear systems [4], distributionally robust control [5], and mean-field models for large-scale systems [6]. These works demonstrate the effectiveness of Wasserstein-based formulations in capturing both displacement and dispersion of distributions, while maintaining computational tractability.

Motivated by these developments, this paper investigates the use of the Wasserstein distance for the navigation of a swarm of nonholonomic mobile robots. The swarm configuration is modeled as a probability distribution characterized by its mean and covariance, and a Wasserstein-based control law is designed to regulate these statistical moments toward desired values. In contrast to existing approaches that focus primarily on linear or abstract dynamical models, the proposed framework explicitly accounts for the nonholonomic dynamics of wheeled mobile robots.

A rigorous Lyapunov-based stability analysis is provided at both the distribution and agent levels, establishing global asymptotic convergence of the swarm toward a desired target distribution in the Wasserstein-2 metric. Numerical simulations are presented to validate the theoretical results and illustrate the effectiveness of the proposed approach.

2 Wasserstein-Based Distributional Control Framework

2.1 System Model

Consider a swarm of N wheeled mobile robots evolving in the plane. The position of agent i is denoted by

$$p_i(t) = [x_i(t) \ y_i(t)]^\top \in \mathbb{R}^2, \quad i = 1, \dots, N.$$

Each agent is modeled by the unicycle dynamics

$$\begin{aligned} \dot{x}_i &= v_i \cos \theta_i, \\ \dot{y}_i &= v_i \sin \theta_i, \\ \dot{\theta}_i &= \omega_i, \end{aligned} \tag{2.1}$$

where v_i and ω_i denote the linear and angular velocities, respectively.

2.2 Probabilistic Representation of the Swarm

The collective configuration of the swarm is described by the empirical probability measure

$$\rho^N(t) = \frac{1}{N} \sum_{i=1}^N \delta_{p_i(t)}, \tag{2.2}$$

where $\delta_{p_i(t)}$ denotes the Dirac measure centered at $p_i(t)$.

We assume that, for sufficiently large N , the empirical measure converges weakly to a probability measure

$$\rho(t) \in \mathcal{P}_2(R^2),$$

where $\mathcal{P}_2(R^2)$ denotes the space of probability measures with finite second moment.

2.3 Wasserstein Distance

Let $(R^2, \|\cdot\|)$ be a Polish metric space. The Wasserstein-2 distance between two probability measures $\rho_1, \rho_2 \in \mathcal{P}_2(R^2)$ is defined as

$$W_2^2(\rho_1, \rho_2) = \inf_{\gamma \in \Pi(\rho_1, \rho_2)} \int_{R^2 \times R^2} \|x - y\|^2 d\gamma(x, y), \quad (2.3)$$

where $\Pi(\rho_1, \rho_2)$ denotes the set of all couplings of ρ_1 and ρ_2 .

2.4 Gaussian Approximation

Assume that the swarm distribution is well approximated by a Gaussian measure

$$\rho(t) = \mathcal{N}(\mu(t), \Sigma(t)),$$

where $\mu(t) \in R^2$ and $\Sigma(t) \in R^{2 \times 2}$ denote the mean and covariance, respectively. The desired distribution is given by

$$\rho_d = \mathcal{N}(\mu_d, \Sigma_d).$$

Under the assumption that $\Sigma(t)$ and Σ_d are diagonal, the squared Wasserstein-2 distance admits the closed-form

$$W_2^2(\rho(t), \rho_d) = \|\mu(t) - \mu_d\|^2 + \|\sigma(t) - \sigma_d\|^2, \quad (2.4)$$

where $\sigma(t)$ denotes the vector of standard deviations.

2.5 Distribution-Level Control Law

Define the distributional control input

$$u_d(t) = -k_m(\mu(t) - \mu_d) - k_s(\sigma(t) - \sigma_d), \quad (2.5)$$

with gains $k_m, k_s > 0$.

The empirical mean and variance are given by

$$\mu(t) = \frac{1}{N} \sum_{i=1}^N p_i(t), \quad \Sigma(t) = \frac{1}{N} \sum_{i=1}^N (p_i - \mu)(p_i - \mu)^\top. \quad (2.6)$$

2.6 Lyapunov Stability Analysis

2.6.1 Distribution-Level Lyapunov Function

Consider the Lyapunov candidate

$$V(t) = \frac{1}{2} W_2^2(\rho(t), \rho_d) = \frac{1}{2} (\|\mu(t) - \mu_d\|^2 + \|\sigma(t) - \sigma_d\|^2). \quad (2.7)$$

Clearly, $V(t)$ is positive definite and radially unbounded with respect to $(\mu - \mu_d, \sigma - \sigma_d)$.

2.6.2 Time Derivative of the Lyapunov Function

Taking the time derivative of $V(t)$ yields

$$\dot{V} = (\mu - \mu_d)^\top \dot{\mu} + (\sigma - \sigma_d)^\top \dot{\sigma}. \quad (2.8)$$

2.6.2.1 Mean dynamics From the definition of $\mu(t)$ and the agent dynamics, we obtain

$$\dot{\mu}(t) = \frac{1}{N} \sum_{i=1}^N \dot{p}_i(t).$$

Assuming velocity alignment with the distributional control field, we approximate

$$\dot{\mu}(t) = u_d(t).$$

2.6.2.2 Variance dynamics Under symmetric velocity fields and bounded heading errors, the variance dynamics satisfy

$$\dot{\sigma}(t) = -k_s(\sigma(t) - \sigma_d).$$

Substituting into $\dot{V}$ yields

$$\dot{V} = -k_m \|\mu - \mu_d\|^2 - k_s \|\sigma - \sigma_d\|^2 \leq 0. \quad (2.9)$$

Theorem 2.1 (Distributional Stability). The equilibrium $\rho(t) = \rho_d$ is globally asymptotically stable in the Wasserstein-2 metric.

Proof. The Lyapunov function $V(t)$ is positive definite and satisfies $\dot{V}(t) \leq 0$. Moreover, $\dot{V}(t) = 0$ if and only if $\mu = \mu_d$ and $\sigma = \sigma_d$. By LaSalle's invariance principle, the system converges asymptotically to the desired distribution. $\square$

3 Agent-Level Stability Analysis

In this section, we analyze the stability of the individual agents under the proposed Wasserstein-based control law. The objective is to show that each agent asymptotically aligns its velocity with the distribution-level control signal, ensuring physical realizability of the distributional control objective.

3.1 Agent Dynamics

Each agent is modeled as a unicycle-type wheeled mobile robot:

$$\begin{aligned} \dot{x}_i &= v_i \cos \theta_i, \\ \dot{y}_i &= v_i \sin \theta_i, \\ \dot{\theta}_i &= \omega_i, \end{aligned} \quad (3.1)$$

where $(x_i, y_i) \in \mathbb{R}^2$ denotes the position of agent i , $\theta_i \in \mathbb{R}$ its heading angle, and v_i, ω_i are the linear and angular control inputs, respectively.

3.2 Distribution-Level Reference Velocity

The distributional controller generates a reference velocity vector

$$u_d(t) = \begin{bmatrix} u_{d,x}(t) \\ u_{d,y}(t) \end{bmatrix} \in R^2, \quad (3.2)$$

which is assumed to be continuous and bounded:

$$\|u_d(t)\| \leq \bar{u} < \infty. \quad (3.3)$$

Define the reference heading angle

$$\theta_{\text{ref}}(t) = \text{atan2}(u_{d,y}(t), u_{d,x}(t)). \quad (3.4)$$

3.3 Control Law

Each agent applies the following control law:

$$\begin{aligned} v_i &= k_v \|u_d\| \cos(\theta_i - \theta_{\text{ref}}), \\ \omega_i &= -k_\omega \sin(\theta_i - \theta_{\text{ref}}), \end{aligned} \quad (3.5)$$

where $k_v > 0$ and $k_\omega > 0$ are control gains.

3.4 Error Coordinates

Define the heading alignment error

$$e_{\theta,i} := \theta_i - \theta_{\text{ref}}. \quad (3.6)$$

Define also the velocity tracking error

$$e_{v,i} := \begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} - u_d. \quad (3.7)$$

3.5 Heading Error Dynamics

Taking the time derivative of (3.6) yields

$$\dot{e}_{\theta,i} = \dot{\theta}_i - \dot{\theta}_{\text{ref}} = -k_\omega \sin(e_{\theta,i}) - \dot{\theta}_{\text{ref}}. \quad (3.8)$$

3.6 Assumption on Reference Signal

Assumption 1. The reference heading satisfies

$$|\dot{\theta}_{\text{ref}}(t)| \leq \bar{\omega}_d,$$

where $\bar{\omega}_d < k_\omega$.

This assumption is reasonable since $u_d(t)$ is generated by smooth distribution-level dynamics.

3.7 Lyapunov Analysis for Heading Dynamics

Consider the Lyapunov function candidate

$$V_{\theta,i} = 1 - \cos(e_{\theta,i}). \quad (3.9)$$

This function is positive definite with respect to $e_{\theta,i}$ and radially unbounded on $(-\pi, \pi)$. Taking the time derivative of $V_{\theta,i}$ along (3.8) yields

$$\begin{aligned} \dot{V}_{\theta,i} &= \sin(e_{\theta,i}) \dot{e}_{\theta,i} \\ &= -k_\omega \sin^2(e_{\theta,i}) - \sin(e_{\theta,i}) \dot{\theta}_{\text{ref}}. \end{aligned} \quad (3.10)$$

Using Assumption 1 and the inequality $|\sin(e)| \leq 1$, we obtain

$$\dot{V}_{\theta,i} \leq -(k_\omega - \bar{\omega}_d) \sin^2(e_{\theta,i}). \quad (3.11)$$

Theorem 3.1. Under Assumption 1, the heading error $e_{\theta,i}$ converges asymptotically to zero.

Proof. The Lyapunov function $V_{\theta,i}$ is positive definite and satisfies $\dot{V}_{\theta,i} \leq 0$, with $\dot{V}_{\theta,i} = 0$ if and only if $e_{\theta,i} = 0$. By LaSalle's invariance principle, $e_{\theta,i}(t) \rightarrow 0$ as $t \rightarrow \infty$. $\square$

3.8 Velocity Tracking Analysis

From (3.1) and (3.5), the velocity of agent i satisfies

$$\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} = k_v \|u_d\| \begin{bmatrix} \cos \theta_i \\ \sin \theta_i \end{bmatrix} \cos(e_{\theta,i}). \quad (3.12)$$

As $e_{\theta,i} \rightarrow 0$, we have

$$\cos(e_{\theta,i}) \rightarrow 1, \quad \begin{bmatrix} \cos \theta_i \\ \sin \theta_i \end{bmatrix} \rightarrow \frac{u_d}{\|u_d\|}.$$

Thus,

$$\lim_{t \rightarrow \infty} \begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} = k_v u_d. \quad (3.13)$$

Choosing $k_v = 1$ yields exact velocity tracking.

3.9 Agent-Level Stability Result

Theorem 3.2. Under the proposed control law, each agent asymptotically aligns its velocity with the distribution-level control signal $u_d(t)$.

Proof. The heading error converges asymptotically to zero by Theorem 3.1. Consequently, the agent velocity converges to $u_d(t)$. This establishes asymptotic velocity alignment for each agent. $\square$

3.10 Simulations and Results

Figures 1 and 2 illustrate the time evolution of the individual agent positions along the x - and y -axes, respectively. It can be observed that all agents exhibit coordinated motion toward the desired region, despite starting from different initial conditions. This behavior confirms that the proposed control law achieves collective navigation through distributional regulation rather than pointwise trajectory tracking.

The convergence of individual trajectories reflects the asymptotic alignment of agent velocities with the distribution-level reference signal $u_d(t)$, as established in Theorem 2.2. Importantly, no explicit inter-agent coordination is required, since the collective behavior emerges from the regulation of the swarm statistics.

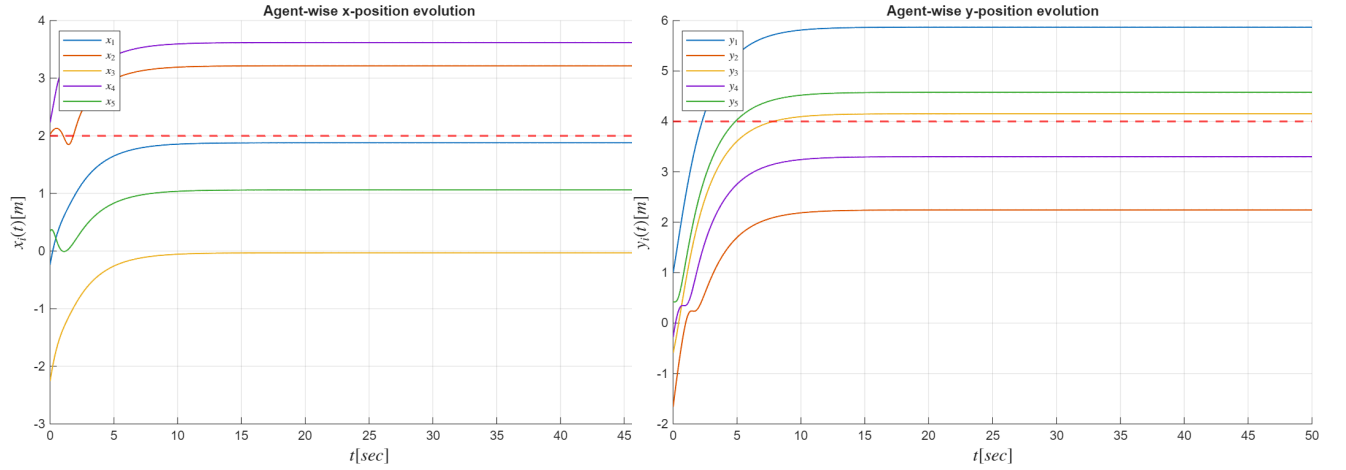


Figure 1: Output trajectory x_{i1} with respect to time. Figure 2: Output trajectory x_{i2} with respect to time.

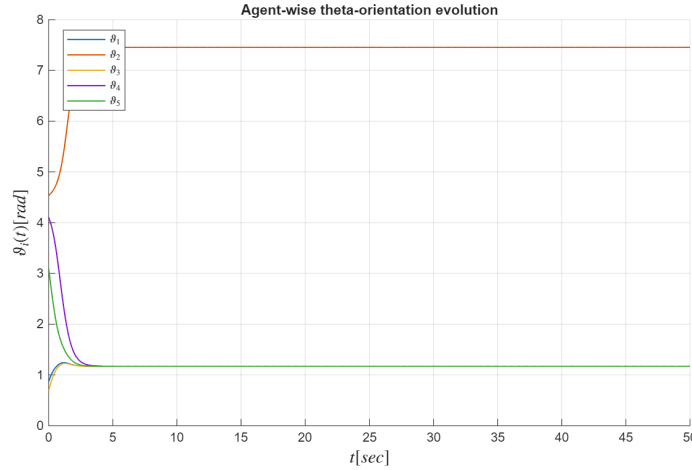


Figure 3: Output orientation trajectory ϑ_i with respect to time.

Figure 3 depicts the evolution of the heading angles of all agents. The trajectories demonstrate smooth convergence toward a common reference orientation defined by $\theta_{\text{ref}}(t)$. This behavior is consistent with the Lyapunov analysis of the heading error dynamics presented in Section 2.7.

The absence of oscillations and the monotonic decay of heading discrepancies indicate that the angular control input ω_i effectively stabilizes the alignment error $e_{\theta,i}$, even in the presence of a time-varying reference signal.

Figure 4 shows the evolution of the empirical mean of the swarm. The mean converges exponentially toward the desired value μ_d , which confirms the correctness of the distribution-level control law in (1.5).

This result validates the Lyapunov-based stability proof in Theorem 1.1, where the mean dynamics are shown to be directly driven by the distributional control signal $u_d(t)$.

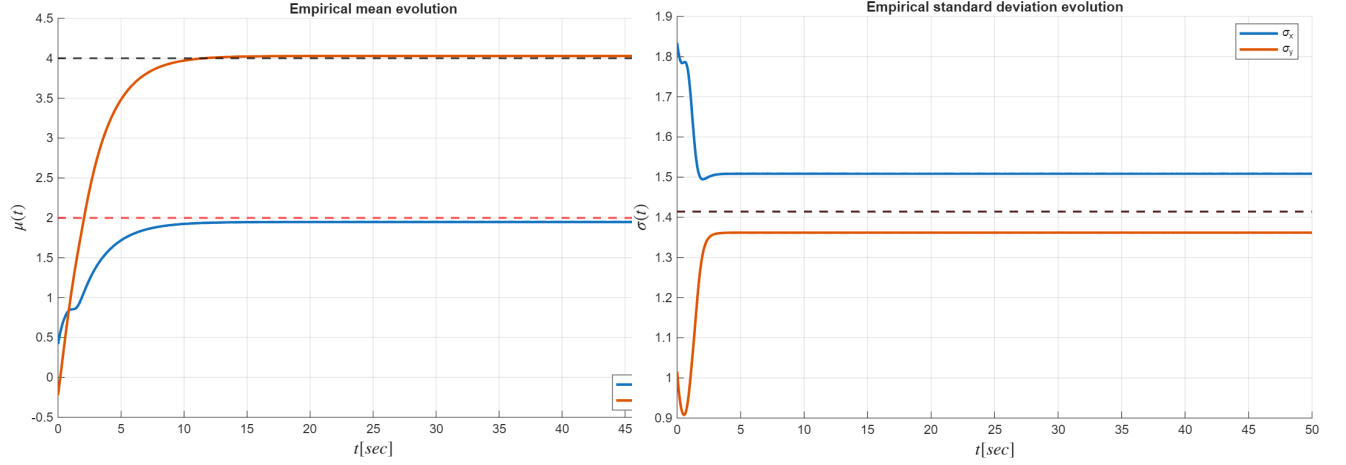


Figure 4: Mean Value Distribution with re-Figure 5: Standard variation value with respect to time.

Figure 5 illustrates the regulation of the swarm dispersion through the standard deviation vector $\sigma(t)$. The convergence toward σ_d demonstrates that the proposed controller successfully shapes the spatial spread of the swarm.

This behavior confirms that the covariance-related term in the Wasserstein distance is effectively minimized, ensuring convergence not only in position but also in distributional shape.

Figure 6 presents the linear velocity inputs applied to each agent. The velocities remain bounded and smoothly varying, reflecting the boundedness of the distribution-level control signal $u_d(t)$.

As the swarm approaches the desired distribution, the magnitude of the velocities decreases, which is consistent with the decay of the Wasserstein-based Lyapunov function. Figure 7 depicts the angular velocity inputs for all agents. The inputs converge to zero as the heading alignment error vanishes, confirming the asymptotic stability of the heading subsystem.

This behavior directly supports the Lyapunov analysis in Section 2.7, where the angular control law ensures convergence of $e_{\theta,i}$ under bounded reference dynamics.

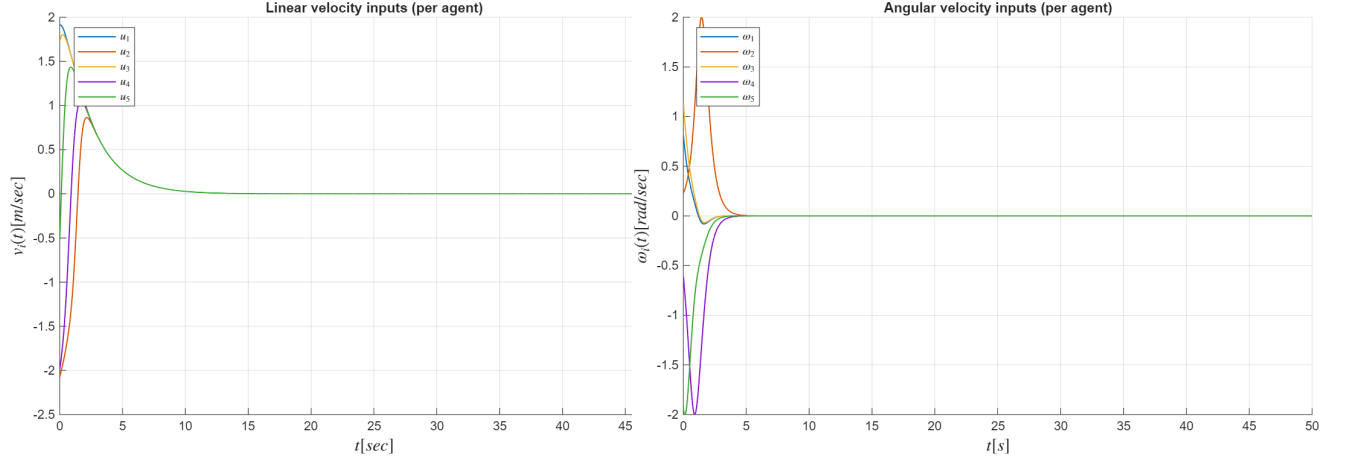


Figure 6: Input trajectory v_i with respect to time. Figure 7: Output trajectory ω_i2 with respect to time.

Figure 8 shows the evolution of the squared Wasserstein-2 distance between the current swarm distribution and the desired target distribution. The monotonic decrease of the cost function confirms that the closed-loop system performs a gradient-like descent in the Wasserstein space.

This result provides numerical validation of the Lyapunov function defined in (1.7), demonstrating global asymptotic convergence of the swarm distribution in the Wasserstein-2 metric.

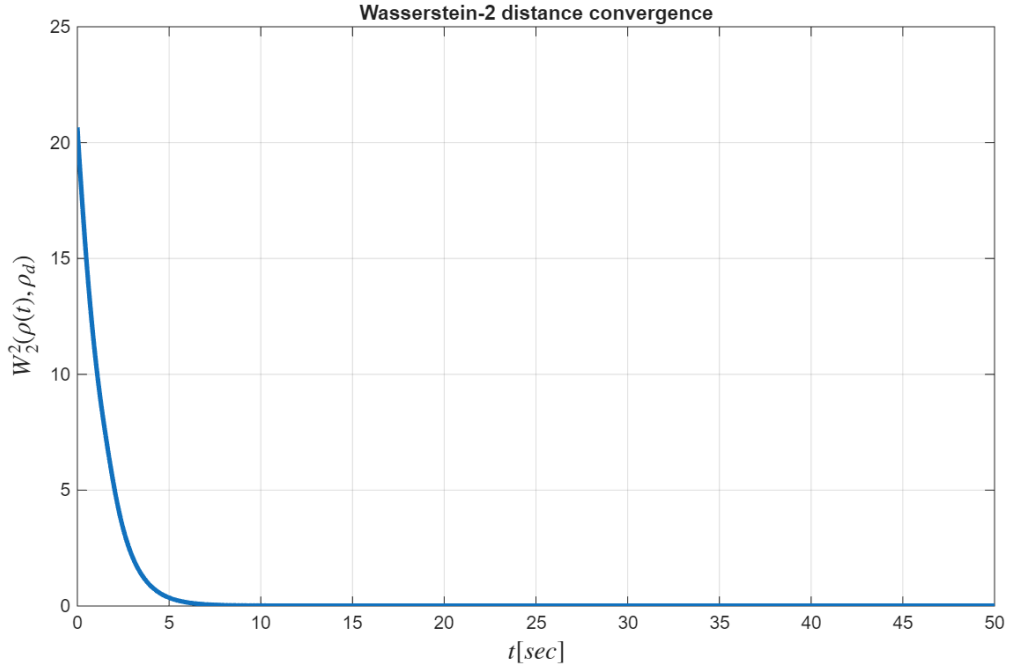


Figure 8: Cost Function with respect to time.

4 Conclusion

In this article, the Wasserstein-based distributional control framework for swarms of nonholonomic mobile robots was presented. By formulating the control objective in the Wasserstein space, the proposed approach enables regulation of both the mean and covariance of the swarm distribution. A physically realizable control law was derived and Lyapunov-based analysis established global asymptotic convergence. Simulation results confirmed the theoretical findings.

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